

MGT411

Solution of Practice Questions

Lessons (1-22)

- 1. How many numbers of prices can be there for 60 commodities under barter system?**

$$\text{Prices} = n(n-1)/2$$

N= numbers of commodities

$$\text{Prices} = 60(60-1)/2$$

$$\text{Prices} = 60 \times 29.5$$

$$\text{Prices} = 1770$$

- 2. Ahmad wants to start his new business 12 years later with Rs. 3,00,000. For this purpose, how much funds he should save and invest today if he gets 5.3% on his funds?**

$$\text{PV} = \text{FV} / (1+i)^n$$

$$\text{PV} = 300,000 / (1.053)^{12}$$

$$\text{PV} = 300,000 / 1.858$$

$$\text{PV} = 161463.93$$

- 3. Suppose you have invested an amount of Rs. 75,000 at 14% interest rate. In how much time this amount will turn to 150,000?**

$$T = 72 / I$$

$$t = 72 / 14$$

$$t = 5.14 \text{ years}$$

- 4. If you have Rs. 50,000 and invest it at 8.5% rate of interest then how much will you get after 5 years?**

$$\text{FV} = \text{PV} \times (1+i)^n$$

$$\text{FV} = 50,000 \times (1+0.085)^5$$

$$\text{FV} = 50,000 \times 1.504$$

$$\text{FV} = 75,182.83$$

5. What will be the yield to maturity (YTM) of 1 year 5% coupon bond having face value \$100 and currently selling at \$ 90?

$$\text{Price} = \{ \text{Coupon payment} / (1 + \text{YTM})^1 \} + \{ \text{Face value} / (1 + \text{YTM})^1 \}$$

$$90 = \{ 5 / (1 + \text{YTM})^1 \} + \{ 100 / (1 + \text{YTM})^1 \}$$

$$1 + \text{YTM} = 105 / 90$$

$$1 + \text{YTM} = 1.166$$

$$\text{YTM} = 1.166 - 1 = 16.7 \%$$

6. You are required to calculate current yield of 6% coupon bond that you have purchased at Rs.107 with face value of Rs. 100.

$$\text{Current Yield} = \text{Yearly Coupon Payment} / \text{Price Paid}$$

$$\text{Current Yield} = 6 / 107$$

$$\text{Current Yield} = 5.61\%$$

7. Use Fisher equation to calculate real interest rate if inflation is 6% and nominal interest rate is 12%.

Fisher equation is:

$$r = i - \pi$$

$$\text{Real interest rate} = \text{nominal interest rate} - \text{inflation}$$

$$\text{Real interest rate} = 12 - 6 = 6\%$$

8. Calculate CPI index for the years 2020, 2021 and 2022 with the given data.

Years	Cost of basket	CPI index
2020	Rs. 1,000	

2021	Rs. 1,500	
2022	Rs. 2,000	

$CPI = (\text{cost of basket in that period} / \text{cost of basket in Base period}) * 100$

Years	Cost of basket	CPI index
2020	1000	100
2021	1500	150
2022	2000	200

9. Calculate inflation rate for the years 2020, 2021 and 2022 with the given data.

Years	Cost of basket	CPI index	Inflation rate
2019	Rs. 800		
2020	Rs. 1,000		
2021	Rs. 1,500		
2022	Rs. 2,000		

$CPI = (\text{cost of basket in that period} / \text{cost of basket in Base period}) * 100$

$\text{Inflation rate} = \{(\text{CPI index that period} - \text{CPI index Proceeding period}) / \text{CPI index Proceeding period}\} * 100$

Years	Cost of basket	CPI index	Inflation rate
2019	Rs. 800	100	-
2020	Rs. 1,000	125	25%
2021	Rs. 1,500	187.5	50%
2022	Rs. 2,000	250	33.33%

10. Inflation can be measured by the Consumer Price Index (CPI). Calculate inflation rate by using CPI values given in following table.

Years	CPI	Inflation rate
2019	100	
2020	110	
2021	115	
2022	126	

Inflation rate = {(CPI index that period – CPI index Proceeding period) / CPI index Proceeding period}*100

Years	CPI	Inflation rate
2019	100	-
2020	110	10%
2021	115	4.5%
2022	126	9.56%

11. Calculate GDP deflator using the data given in following table.

Years	Nominal GDP (Rs.)	Real GDP (Rs.)	GDP deflator
2020	65,000	62,100	
2021	70,500	68,000	
2022	82,500	80,200	

GDP Deflator = Nominal GDP/Real GDP * 100

Years	Nominal GDP (Rs.)	Real GDP (Rs.)	GDP deflator
2020	65,000	62,100	104.67
2021	70,500	68,000	103.68
2022	82,500	80,200	105.98

12. Consider an investment of value Rs. 1,000 having an equal probability of increase in value to Rs, 1,200 and fall to Rs. 800. You are required to calculate its Expected value.

Possibilities	Probability	Pay off	Payoff * probability
1	1/2	800	400
2	1/2	1200	600

Expected value = Sum of (Probability times payoff) = \$1000

13. Consider an investment of value Rs. 1,000 having an equal probability of fall and rise. You are required to calculate its variance, and standard deviation.

Possibilities	Probability	Pay off
1	1/2	600
2	1/2	1400

Possibilities	Probability	Pay off	Payoff * probability
1	1/2	600	300
2	1/2	1400	700

Expected value = Sum of (Probability times payoff) = \$1000

$$\text{Variance} = \sum P[X - \sum E(X)]^2$$

$$\text{Variance} = \frac{1}{2} [(600 - 1000)^2 + (1400 - 1000)^2]$$

$$\text{Variance} = \frac{1}{2} [(-400)^2 + (400)^2]$$

$$\text{Variance} = \frac{1}{2} [160000 + 160000]$$

$$\text{Variance} = \frac{1}{2} [320000]$$

$$\text{Variance} = 160000$$

Standard deviation = Square root of variance

$$\text{Standard deviation} = 400$$

14. Mr. Ali wants to have the amount of Rs. 12,00,000 fifteen years from now as savings. If he earns 10.5% rather than 9% then how much less he must deposit today to have this amount 15 years later?

$$PV = FV / (1+i)^n$$

$$PV = 12,00,000 / (1.09)^{15} = 307613.43$$

$$PV = 12,00,000 / (1.105)^{15} = 268396.33$$

$$\text{Difference} = 39217$$

15. Suppose you purchased a 6% coupon bond for 10 years having a face value Rs. 100 at the cost of Rs. 100. What will be the holding period returns if you sell this bond in the market a year later for Rs. 110?

Holding Period Return = (Yearly Coupon payment/Bond price paid) + (Selling price – face value/Price of the bond)

$$= (6/100) + (110-100/100)$$

$$= 0.06 + 0.1$$

$$= 0.16 \text{ or } 16\%$$

16. Calculate the yield of a tax-free bond if taxable bond yield is 12% and income tax rate is 30%.

$$\text{Tax-Exempt Bond Yield} = (\text{Taxable Bond Yield}) \times (1 - \text{Tax Rate})$$

$$= 0.12 \times (1 - 0.3)$$

$$= 0.12 \times 0.7$$

$$= 0.084 = 8.4\%$$

17. Calculate the market price of a 7% coupon bond that matures in 3 years with 8.5% yield to maturity. Face value of the bond is Rs. 100.

$$\text{Price} = 7/(1.085)^1 + 7/(1.085)^2 + 7/(1.085)^3 + 100/(1.085)^3$$

$$= 6.45 + 5.94 + 5.48 + 78.30$$

$$= 96.17$$